

1 Fubini's Theorem

For a rectangle $Q = [a, b] \times [c, d] \subseteq \mathbb{R}^2$, let $f : Q \rightarrow \mathbb{R}$ be continuous. Then, for every fixed $x \in [a, b]$, the map $y \mapsto f(x, y)$ is continuous on $[c, d]$, and hence Riemann integrable. We define

$$g(x) := \int_c^d f(x, y) \, dy.$$

Then the *iterated integral* is

$$\int_a^b g(x) \, dx = \int_a^b \int_c^d f(x, y) \, dy \, dx.$$

Moreover, by Fubini's theorem, we can interchange the order,

$$\int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy = \int_Q f.$$

Theorem 1.1 (Fubini). Let $f : Q \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, with $Q = A \times B$, where $A \subseteq \mathbb{R}^k$ and $B \subseteq \mathbb{R}^{n-k}$ are rectangles. Suppose f is Riemann integrable on Q . For each fixed $x \in A$, define the section

$$f_x : B \rightarrow \mathbb{R}, \quad f_x(y) := f(x, y).$$

Then f_x is Riemann integrable on B , and the function

$$g : A \rightarrow \mathbb{R}, \quad g(x) := \int_B f_x(y) \, dy$$

is Riemann integrable on A . Moreover,

$$\int_Q f = \int_A \left(\int_B f_x(y) \, dy \right) dx.$$

Proof. Let P be a partition of Q . Since $Q = A \times B$ the partition P induces partitions P_A of A and P_B of B and, $P = P_A \times P_B$.

Correspondingly, for $R_A \in P_A$ and $R_B \in P_B$, we have the subrectangle $R_{A \times B} := R_A \times R_B$.

Fix $R_A \in P_A$. For any $x \in R_A$ and $y \in R_B$, we have

$$m_{R_A \times R_B}(f) \leq f(x, y),$$

where

$$m_{R_A \times R_B}(f) := \inf_{(u, v) \in R_A \times R_B} f(u, v).$$

Multiplying by $\text{vol}(R_B)$ and summing over all $R_B \in P_B$ gives

$$\sum_{R_B \in P_B} m_{R_A \times R_B}(f) \, \text{vol}(R_B) \leq L(f_x, P_B),$$

where

$$L(f_x, P_B) := \sum_{R_B \in P_B} m_{R_B}(f_x) \, \text{vol}(R_B)$$

is the lower Darboux sum of the section $f_x(y) = f(x, y)$ over B . From the preceding estimates, we have

$$L(f, P) \leq \sum_{R_A \in P_A} L(f_x, P_B) \, \text{vol}(R_A), \quad U(f, P) \geq \sum_{R_A \in P_A} U(f_x, P_B) \, \text{vol}(R_A)$$

Now define

$$g : A \rightarrow \mathbb{R}, \quad g(x) := \int_B f(x, y) \, dy = \int_B f_x(y) \, dy.$$

For each $R_A \in P_A$, let

$$m_{R_A}(g) := \inf_{x \in R_A} g(x), \quad M_{R_A}(g) := \sup_{x \in R_A} g(x).$$

Thus

$$L(g, P_A) = \sum_{R_A \in P_A} m_{R_A}(g) \, \text{vol}(R_A), \quad U(g, P_A) = \sum_{R_A \in P_A} M_{R_A}(g) \, \text{vol}(R_A)$$

For any $x_A \in R_A$,

$$L(f_{x_A}, P_B) \leq \int_B f_{x_A} \, dy = g(x_A).$$

Hence

$$L(f_{x_A}, P_B) \leq g(x_A) \leq U(f_{x_A}, P_B).$$

Taking the infimum and supremum over $x_A \in R_A$,

$$\inf_{x_A \in R_A} L(f_{x_A}, P_B) \leq m_{R_A}(g) \leq M_{R_A}(g) \leq \sup_{x_A \in R_A} U(f_{x_A}, P_B).$$

In particular, using $L(g, P_A) \leq U(g, P_A)$, together with the estimates above, we obtain

$$L(f, P) \leq L(g, P_A) \leq U(g, P_A) \leq U(f, P).$$

Therefore,

$$\boxed{U(g, P_A) - L(g, P_A) \leq U(f, P) - L(f, P)}.$$

Since f is Riemann integrable, given $\epsilon > 0$, there exists a partition P of Q such that

$$U(f, P) - L(f, P) < \epsilon.$$

For the induced partition P_A of A ,

$$U(g, P_A) - L(g, P_A) < \epsilon.$$

Hence, by the criterion for Riemann integrability, g is Riemann integrable on A . Moreover,

$$L(f, P) \leq L(g, P_A) \leq \int_A g \leq U(g, P_A) \leq U(f, P).$$

Thus

$$\left| \int_A g - \int_Q f \right| \leq U(f, P) - L(f, P).$$

Since the right-hand side can be made arbitrarily small,

$$\boxed{\int_A g = \int_Q f}.$$

□

The same argument may be iterated for $Q = A \times B \subseteq \mathbb{R}^n$ splitting the variables successively into blocks. This yields the corresponding result for iterated integrals in several variables.